සියලු ම හිමිකම් ඇවිරිණි / முழுப் பதிப்புரிமையுடையது / All Rights Reserved]

නව/පැරණි නිර්දේශය – புதிய/பழைய பாடத்திட்டம் – New/Old Syllabus

NEW/OLD

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2020
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2020
General Certificate of Education (Adv. Level) Examination, 2020

කෘෂි තාක්ෂණවේදය I விவசாயத் தொழினுட்பவியல் I **Agro Technology I**	**18 E I**	පැය දෙකයි இரண்டு மணித்தியாலம் ***Two hours***

Instructions:

* *Answer* ***all*** *questions.*
* *Write your* ***Index Number*** *in the space provided in the answer sheet.*
* *Instructions are given on the back of the answer sheet. Follow those carefully.*
* *In each of the questions* **1** *to* **50**, *pick one of the alternatives from* (1), (2), (3), (4), (5) *which is* ***correct*** *or* ***most appropriate*** *and* ***mark your response on the answer sheet with a cross*** (×) *on the number of the correct option in accordance with the instructions given on the back of the answer sheet.*

1. The major field of soft technology is
(1) Nanotechnology. (2) Biotechnology.
(3) Industrial technology. (4) Food technology.
(5) Information technology.

2. The major atmospheric pollutant that is generated from agricultural activities is
(1) CH_4 (2) CO_2 (3) NO_2 (4) N_2O (5) CFC

3. The basement of the food pyramid represents the required amount of
(1) fat for a balanced diet. (2) proteins for a balanced diet.
(3) vitamins for a balanced diet. (4) minerals for a balanced diet.
(5) carbohydrates for a balanced diet.

4. Select the correct formula that is used to calculate the Body Mass Index (BMI) of an individual.

(1) $\text{BMI} = \dfrac{\text{Height of a person (cm)}}{[\text{Weight of the person (kg)}]^2}$

(2) $\text{BMI} = \dfrac{\text{Height of a person (cm)}}{\text{Weight of the person (kg)}}$

(3) $\text{BMI} = \dfrac{\text{Weight of a person (kg)}}{[\text{Height of the person (m)}]^2}$

(4) $\text{BMI} = \dfrac{\text{Weight of a person (kg)}}{\text{Height of the person (m)}}$

(5) $\text{BMI} = \dfrac{\text{Weight of a person (kg)}}{[\text{Height of the person (cm)}]^2}$

5. The macro nutrients in food mainly contribute to
(1) prevention from diseases. (2) successful reproduction.
(3) growth and development. (4) maintaining a healthy body.
(5) improve brain function.

[see page two

6. Consider the following statements.

A - Oxidation of lipids can be explained as a non-enzymatic process.

B - Peroxides are the final products generated from lipid oxidation.

C - Photo-oxidation of lipid can be controlled by using carotenoids.

Of the above, the correct statement/s is/are

(1) A only. (2) B only. (3) C only.

(4) A and B only. (5) A and C only.

7. The non-organoleptic parameter which can change due to spoilage of food is

(1) taste. (2) colour. (3) texture. (4) odour. (5) mineral content.

8. Food security and food safety are best explained as,

(1) accessibility of food with no health hazards and adequate food, respectively.

(2) availability of adequate food and food with moderate health hazards, respectively.

(3) accessibility for non-hazardous food and adequate food, respectively.

(4) availability of nutritious food and its ability to safeguard human health, respectively.

(5) availability of adequate food with correct nutrition and no health hazards, respectively.

9. Consider the following statements.

A - Use of cold chain to distribute fruits and vegetables may assure food security at the national level.

B - Practice of cold chain may extend the shelf-life of fruits and vegetables.

Of the above,

(1) Both A and B are correct.

(2) A is correct and B is incorrect

(3) A is incorrect and B is correct.

(4) A is correct and it is further explained by B.

(5) B is correct and it is further explained by A.

10. A farmer transplanted chilli seedlings obtained from a nursery bed and covered them with coconut leaves. The purpose of providing cover for the seedlings after transplanting is to protect them from

(1) wind. (2) rain. (3) insects. (4) frost. (5) direct sun light.

11. The environmental parameter/s, which influence the soil genesis is/are

(1) humidity. (2) temperature. (3) rainfall.

(4) humidity and temperature. (5) rainfall and temperature.

12. Consider the following statements.

A - Seed dormancy would help seeds to overcome unfavourable conditions.

B - Scarification of passion fruit (*Passiflora edulis*) seeds helps removing seed dormancy.

C - Mature viable crop seeds have a dormancy period of more than 2 weeks.

Of the above, the correct statement/s is/are

(1) A only. (2) B only. (3) C only.

(4) A and B only. (5) B and C only.

13. An advantage of row seeding as a method of field establishment of crop seeds is the

(1) facilitation of removing unhealthy seedlings.

(2) facilitation of weed control using weeding equipment.

(3) ability to provide optimum conditions for seed germination.

(4) ability to transplant excess seedlings in the extra spaces available in rows.

(5) ability to obtain vigorous seedlings compared to other crop establishment methods.

[see page three

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14. Seed purity percentage is a main factor that should be considered in the seed industry. Presence of seeds belonging to the paddy variety Bg358 and the weed *Echinochloa crus-galli* in a seed lot of the paddy variety Bg360, could be best described as an issue related to

(1) physical purity only. (2) genetic purity only.
(3) species and physical purity only. (4) genetic and physical purity only.
(5) weed and physical purity only.

15. Kothalahimbutu (*Salacia reticulata*) is used to treat

(1) asthma. (2) chicken-pox. (3) measles. (4) Mumps. (5) diabetes.

16. Select the medicinal plant that the roots are used for de-coxing mixtures.

(1) Bulu (2) Aralu (3) Ginger (4) Nelli (5) Adathoda

17. From the following, select the most appropriate tool to obtain the economically important harvest of Savandara for medicinal purposes.

(1) Knife (2) Pair of Scissors (3) Spoon
(4) Scraper (5) Crow bar

18. Select the fish that needs a brackish water environment to complete it's life cycle.

(1) Snake-head fish (Loola) (2) Cat fish (Magura) (3) Carp
(4) Tilapia (5) Wekkaya

19. The most appropriate example to describe a method of lowering post-harvest losses is

(1) making curd from milk.
(2) wrapping of papaya fruits using paper during transportation.
(3) use of rice bran as an animal feed.
(4) feeding of leftover food to domestic animals.
(5) application of bio-pesticides to control leafhopper.

20. The post-harvest losses of fruits and vegetables in Sri Lanka is generally considered to be

(1) 5%-10%. (2) 10%-20%. (3) 20%-40%. (4) 40%-60%. (5) 60%-80%.

21. The foundation unit of a living organism is

(1) tissue. (2) protein. (3) cell. (4) organelle. (5) nucleic acid.

22. A fresh graduate in management studies plans to start a poultry farm. A suitable land and a market with high demand are available for this project, but several companies supply chicken meat for this market. According to the SWOT analysis the strength, weakness, opportunity and threat of this business are

(1) suitable land, insufficient training, market and other suppliers, respectively.
(2) market, suitable land, insufficient training and other suppliers, respectively.
(3) suitable land, management degree, market and insufficient training.
(4) other suppliers, management degree, insufficient training and market.
(5) market, other suppliers, insufficient training and management degree.

23. An example for a production-oriented food based technological entrepreneurship is

(1) food catering service.
(2) operating a restaurant.
(3) operating a dairy cattle farm.
(4) operating a yoghurt factory.
(5) selling face masks to get protected from COVID - 19 disease.

[see page four

24. Select the correct statement regarding composting.

(1) The C:N ratio declines with the composting of the raw material.
(2) The release of CO_2 is more at the end of composting.
(3) The C content increases in the pile during composting.
(4) There is a continuous increase of temperature of the pile during composting.
(5) The moisture content in the compost pile increases at the end of composting.

25. Different production processes could be explained using standard symbols. The correct procedure to represent the manufacturing process of set-yoghurt using standard symbols is

(1) ○ → △ → □ → ▽
(2) △ → ○ → □ → ▽
(3) □ → △ → ○ → ▽
(4) ○ → □ → △ → ▽
(5) □ → ○ → △ → ▽

26. Consider the following statements.

A - Hardware infrastructures are essential to use software
B - Internet is required to use software
C - Use of commercial software without licence is an offence

Of the above, the use of software in technological applications is best described by

(1) A only. (2) B only. (3) A and B only.
(4) A and C only. (5) B and C only.

27. Ploughs are normally used for

(1) digging ditches. (2) inter-cultivation. (3) making plant beds.
(4) primary land preparation. (5) secondary land preparation in low lands.

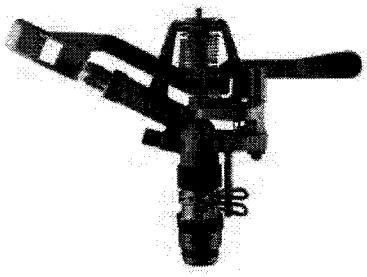
28. When using mechanical seeders, the plant spacing in the row is controlled by

(1) the seed tube. (2) the furrow opener. (3) pressing wheel.
(4) the forward speed. (5) seed metering device.

29. Pitcher irrigation is more suitable for

(1) greenhouses in the up-country of Sri Lanka.
(2) coconut grown in sandy soils in Sri Lanka.
(3) vegetables grown in sandy soils in Sri Lanka.
(4) field crops in the wet zone of Sri Lanka.
(5) orchards in the dry zone of Sri Lanka.

30. This equipment is a part of



(1) a plough.
(2) a knapsack sprayer.
(3) an irrigation system.
(4) a land preparation equipment.
(5) a plant protection equipment.

[see page five

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Past Papers Wiki

31. An example for subsurface irrigation is

(1) deep open ditch. (2) drip irrigation. (3) furrow irrigation.
(4) ring irrigation. (5) basin irrigation.

32. Consider the following condition.

"Leaf mottling, wilting of tip of the leaf blade and chlorosis in mature leaves were observed in maize plants".

The above condition could be properly described as

(1) deficiency of Cl as Cl is a mobile micro element.
(2) deficiency of Zn as Zn is a mobile micro element.
(3) deficiency of Cl as Cl is a non-mobile micro element.
(4) deficiency of Zn as Zn is a non-mobile micro element.
(5) deficiency of Cu as Cu is a non-mobile micro element.

33. Consider the following statements regarding Phospho-compost.

A - Phosphorous-enriched compost made by mixing 10% w/w of rock phosphate with goat manure and pig manure.
B - Phosphorous-enriched compost made by mixing 15% w/w of rock phosphate with pig manure.
C - Phosphorous-enriched compost made by mixing 5% w/w of rock phosphate with goat manure, pig manure and cattle manure.

Of the above, the correct statement/s is/are

(1) A only. (2) B only. (3) C only.
(4) A and B only. (5) B and C only.

34. Improving the use-efficiency of agricultural inputs is one of the major ways to increase crop productivity. In this respect, select the **most correct** statement from the following.

(1) Triple super phosphate is an essential component of basal fertilizer in paddy cultivation in Sri Lanka.
(2) Application of the total requirement of Muriate of Potash as a basal dressing will increase the uptake of K_2O by plants.
(3) Nitrogen losses from urea can be minimized by single application of the total nitrogen requirement of the crop as a top dressing.
(4) Application of slow release fertilizer as a basal dressing will provide adequate nutrients to crops after flowering.
(5) Nano-technology can be effectively used to improve timely and continuous release of nutrients from fertilizer.

35. The hormone that is responsible for breaking dormancy in seeds is

(1) Ethylene. (2) Gibberellins. (3) Cytokinin.
(4) Abscisic acid. (5) Indoleacetic acid.

36. Select the correct statement from the following regarding plant tissue culture.

(1) Formation of callus tissue is called organogenesis.
(2) Potato dextrose is the most suitable gelling agent.
(3) Mercuric bromide is used for surface sterilization of ex-plants.
(4) Proteolytic enzymes help production of protoplasts from the callus tissue.
(5) Serum albumin is a basic component used in culture media.

[see page six

37. Pre-emergence herbicides are used to
(1) destroy weed seedlings when they are at 2-3 leaf stage.
(2) kill above ground vegetative organs.
(3) control weeds in non-agricultural land.
(4) control weeds before planting the crop.
(5) control propagules of weeds found in soil.

38. The most effective preventive technique to control brown plant hopper attack in paddy is
(1) deep ploughing. (2) continuous flooding.
(3) repeated use of pesticides. (4) scattered planting of rice in the area.
(5) increase seed paddy rates for planting.

39. Select the correct statement on pest and disease management of crops in Sri Lanka.
(1) Virus diseases are the most common category of crop diseases.
(2) Leaf curl disease is the most devastating disease of chilli.
(3) Copper sulphate is the most widely used fungicide.
(4) Soil sterilization in plant nursery controls majority of the insect pests of crops.
(5) Flooding is the most effective post-planting technique to control pests and diseases in rice.

40. The most important Personal Protection Equipment (PPE) to be used in pesticide application is
(1) reading the label.
(2) wearing gloves.
(3) using an appropriate spraying equipment.
(4) not to spray during strong winds.
(5) covering the whole body with suitable clothing.

41. The protected culture structure shown in the diagram having a polythene canopy and open sides is a
(1) greenhouse.
(2) Lath house.
(3) solar propagator.
(4) rain shelter.
(5) polytunnel.



42. One of the major limiting factors for using greenhouses for agricultural production in Sri Lanka is the
(1) strong wind. (2) high temperature. (3) high rainfall.
(4) high capital cost. (5) low humidity at night.

43. The correct procedure that should be followed to prepare Gerbera cut flowers, after harvesting and removing the woody basement part, for the export market is
(1) packing individual flowers vertically in shallow cardboard containers, dipping flower stems in 100 ppm silver nitrate solution and storage at 1°C.
(2) dipping flower stems in 100 ppm silver nitrate solution, packing individual flowers horizontally in shallow cardboard containers and storage at 1°C.
(3) dipping flower stems in 40 ppm sodium hypochlorite solution, packing individual flowers horizontally in shallow cardboard containers and storage at 1°C.
(4) dipping flower stems in 100 ppm silver nitrate solution, packing individual flowers horizontally in shallow cardboard containers and storage at 10°C.
(5) dipping flower stems in 100 ppm sodium hypochlorite solution, packing individual flowers vertically in shallow cardboard containers and storage at 1°C.

[see page seven

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44. Consider the following statements which were described the best stage for harvesting roses as cut flowers.

A - Open-bud stage when the colour is fully developed with 1-2 mature leaves.
B - Tight-bud stage when the colour is fully developed with 1-2 mature leaves.
C - Tight-bud stage when the colour is fully developed with 3-4 mature leaves.

Of the above, the correct statement/s is/are

(1) A only. (2) B only. (3) C only.
(4) A and B only. (5) B and C only.

45. One of the objectives of using border plants is

(1) weed control. (2) to fill the free space.
(3) to attract animals. (4) moisture conservation.
(5) to cover unwanted views.

46. The main nutrient found in maize seeds is

(1) lipids. (2) fibre. (3) starch. (4) proteins. (5) vitamins.

47. Following are some statements on formulation of animal feeds.

A - Compared to broiler feeds, layer feeds should contain more calcium and less protein.
B - Compared to milking cow feeds, heifer feeds should contain more protein and calcium.
C - Among all poultry feeds, the highest protein content is found in feed given to broiler chicks.
D - Pigments are usually added to all types of poultry feeds.

Of the above, the correct statements are

(1) A and B only. (2) A and C only. (3) B and C only.
(4) B and D only. (5) C and D only.

48. Candling during incubation is carried out

(1) to observe the position of the embryo.
(2) for sexing of chicks.
(3) to calculate the hatching percentage.
(4) after fifteen days of incubation.
(5) to remove the unsuitable eggs from the incubator.

49. Following statements are on artificial insemination of cattle.

A - Transmission of sexual diseases will be reduced due to artificial insemination.
B - The objective of semen dilution is to preserve semen for future use.
C - Sperm concentration of semen can be estimated by visual test.

Of the above,

(1) Only A is correct. (2) Only B is correct. (3) Only C is correct.
(4) Both A and B are correct. (5) Both B and C are correct.

50. Of the following, select the **most correct** statement.

(1) Fresh milk can be kept for about two weeks under refrigeration.
(2) Milk is preserved to enhance its nutritional quality.
(3) Sterilized milk can be kept for a longer period than pasteurized milk.
(4) Pasteurized milk can be kept for a longer period than sterilized milk.
(5) Both pasteurization and sterilization of milk could kill all micro-organisms in milk.

* * *

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සියලු ම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved]

නව/පැරණි නිර්දේශය – புதிய/பழைய பாடத்திட்டம் – New/Old Syllabus

NEW/OLD

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2020
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2020
General Certificate of Education (Adv. Level) Examination, 2020

කෘෂි තාක්ෂණවේදය விவசாயத் தொழினுட்பவியல் **Agro Technology**	**II** **II** **II**

18 E II

පැය තුනයි மூன்று மணித்தியாலம் ***Three hours***	අමතර කියවීම් කාලය - මිනිත්තු 10 යි மேலதிக வாசிப்பு நேரம் - 10 நிமிடங்கள் **Additional Reading Time - 10 minutes**

Use additional reading time to go through the question paper, select the questions you will answer and decide which of them you will prioritise.

Index No. : .....

Instructions:

* *This question paper consists of* **10** *questions in* **09** *pages.*
* *This question paper comprises of* Parts **A**, **B** *and* **C**.

  The time allotted for all ***three*** *parts is* ***three hours***.

Part A - Structured Essay (02-08 pages)

* *Answer* ***all*** *questions on* ***this paper itself***.
* *Write your answers in the space provided for each question. Please note that the space provided is sufficient for your answers and extensive answers are* ***not*** *expected.*

Part B and C - Essay : (page No. 09)

* *Select* ***two*** *questions from each of the* Parts **B** *and* **C** *and answer* ***four*** *questions only. Use the papers supplied for this purpose.*
* *At the end of the time allotted for this paper, tie the* ***three*** *parts together so that* Part **A** *is on the top of* Parts **B** *and* **C** *before handing over to the supervisor.*
* *You are permitted to remove only* Parts **B** *and* **C** *of the question paper from the Examination Hall.*

For Examiner's Use only

Part	Question No.	Marks
A	1	
	2	
	3	
	4	
B	5	
	6	
	7	
C	8	
	9	
	10	
Total		

Total

In Numbers	
In Letters	

Code Numbers

Marking Examiner 1.	
Marking Examiner 2.	
Marks Checked by	
Supervised by	

[see page two

Part A - Structured Essay

Answer ***all questions on this paper itself.***

(Each question carries **100** *marks.)*

Do not write in this column

1. (A) State the **two** controversial concepts that describe how human perceive the interaction among them.

(1) .....

(2) .....

(B) State a method each to control pollution from following activities.

(1) Particulate emission due to burning diesel in agricultural tractors:

.....

.....

(2) Runoff of excess nutrients from farmlands to surface water:

.....

.....

(C) State **two** advantages of providing a balanced diet to school children.

(1) .....

.....

(2) .....

.....

(D) Write **two** disadvantages of frequent consumption of fast foods.

(1) .....

(2) .....

(E) Spoilage of food could occur due to different reasons. State **one** reason for each of the following food spoilage conditions.

Spoilage condition	Reason
(1) Rancidity	.....
(2) Formation of brown colour of certain cut fruits	.....
(3) Development of sour taste in milk products	.....
(4) Development of black spots on bread slices	.....

(F) Propose **two** ways to achieve household food security.

(1) .....

(2) .....

[see page three

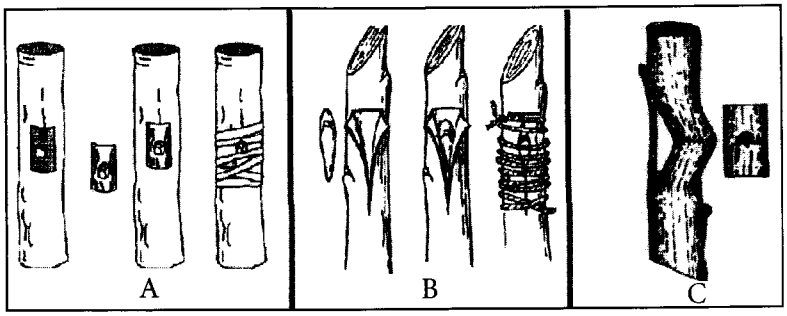
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(G) Names of species of few commonly used agricultural crops are given below. State a vegetative propagation structure found in each species.

Species	Vegetative propagation structure
(1) *Centella asiatica*	.....
(2) *Solnaum tuberosum*	.....

(H) Name the different methods of budding shown in the diagram given below.



A .....

B .....

C .....

(I) Write the appropriate propagule and a plant part used as medicine in each of the following medicinal plants.

Medicinal plant	Propagule	A part used for medicine
(1) Aratta	.....	(1).....
(2) Lunuwila	.....	(2).....
(3) Ginger	.....	(3).....

Q. 1 100

2. (A) (1) Name months in the year in which the highest rainfall is expected by the farmers in the Dry Zone of Sri Lanka.

.....

(2) Name an instrument that is used to measure the speed of the wind.

.....

(3) List **two** factors to be considered in establishing a rain gauge in a meteorological station.

(i) .....

(ii) .....

(B) Write **one** example each for the brackish and salt water ecosystems.

Ecosystem	Example
Brackish water ecosystem	.....
Salt water ecosystem	.....

[see page four

(C) State **four** methods to minimize post-harvest losses in vegetables.

(1) .....

(2) .....

(3) .....

(4) .....

(D) List **two** examples for traditional biotechnological techniques used in agriculture.

(1) .....

(2) .....

(E) List **two** good characteristics of a successful entrepreneur.

(1) .....

(2) .....

(F) Following are few incidences that happened to a business involved in producing herbal tea. State whether they are changes occured in the internal environment or external environment of the business.

(1) Drop in production due to low supply of raw materials.

.....

(2) Reduction in quality of tea due to errors in the production process.

.....

(G) State **two** reasons for preparing a market plan before starting a food production business.

(1) .....

(2) .....

(H) State **two** components of a cash flow statement.

(1) .....

(2) .....

(I) State **three** risks associated with using soft technology.

(1) .....

(2) .....

(3) .....

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[see page five

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(J) Pile method and pit method are the widely used composting methods. Name **two** other methods of composting practiced in Sri Lanka.

(1) .....

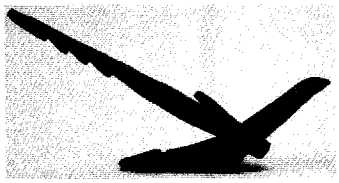
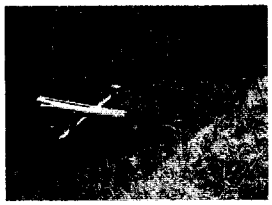
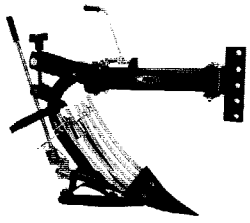
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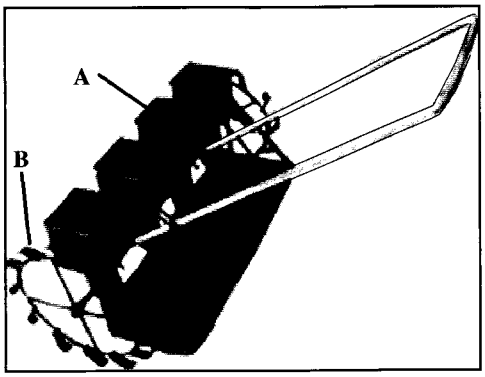
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3. (A) Write **one** use each of the following implement/equipment in agriculture.

Implement/equipment	Use	
	(1)	....
	(2)	....
	(3)	....
	(4)	....

(B) The diagram given below is an agricultural implement.



(1) Write the use of the above implement.

.....

.....

[see page six

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(2) Explain the working principle of the given implement.

.....

.....

.....

.....

(3) Name the parts A and B of the implement and state the function of each part.

Part	Name	Function
A	.....	.....
B	.....	.....

(C) List **three** types of water lifting devices/methods used in agriculture and write **one** advantage of each.

	Water lifting device/method	Advantage
(1)	.....	.....
(2)	.....	.....
(3)	.....	.....

(D) List **three** measures/activities of conserving irrigation water.

(1) .....

(2) .....

(3) .....

(E) List **two** main forms that phosphorus is absorbed by plants.

(1) .....

(2) .....

(F) State the major nutrient and its content (as a percentage) in each of the following fertilizers.

Fertilizer	Major nutrient	Percentage of the major nutrient
Muriate of potash	(1).....	(1).....
Ammonium sulphate	(2).....	(2).....

[see page seven

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(G) (1) Define photosynthesis

.....

.....

(2) State **two** impacts of increasing environmental temperature on metabolism in plants.

(i) .....

(ii) .....

(H) State **two disadvantages** of using tissue cultured plants in agriculture.

(1) .....

(2) .....

Do not write in this column

Q. 3

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4. (A) (1) State **two** advantages and **two** disadvantages of using biocontrol agents for pest management.

Advantages

(i) .....

(ii) .....

Disadvantages

(i) .....

(ii) .....

(2) State **three** advantages of using pesticides in crop production.

(i).....

(ii).....

(iii).....

(B) Name **two** chemical parameters that should be continuously monitored in a growth medium used in protected culture to support growth of a plant.

(1) .....

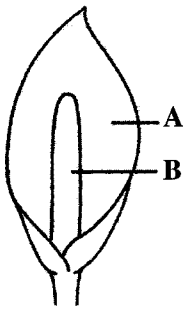
(2) .....

(C) (1) State **one** major propagating material that can be used for each of the following cut-flower types.

Cut flower type	**Major propagating material**
i. Orchid	.....
ii. Rose	.....

[see page eight

(2) Label the parts **A** and **B** of an Anthurium inflorescence.



A : .....

B : .....

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(D) Name **two** hard structures used in landscape architecture.

(1) .....

(2) .....

(E) Name an example of a feedstuff for each of the following feed categories.

(1) Animal protein supplement : .....

(2) Energy supplement : .....

(F) State whether the following grass are fodder or pasture grasses.

(1) Napier grass : .....

(2) Brachiaria grass : .....

(G) List **two** external qualities of eggs suitable for incubation.

(1) .....

(2) .....

(H) Name **two** hormones that are used in the embryo transfer process.

(1) .....

(2) .....

(I) Meat can be preserved by smoking and salting. Explain the main reason for preservation of meat by these two methods.

(1) Smoking

.....

.....

(2) Salting

.....

.....

Q. 4

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NEW/OLD

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இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2020
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2020
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விவசாயத் தொழினுட்பவியல் II
Agro Technology II

18 E II

Essay

* *Select* ***two*** *questions from each of the* Parts **B** *and* **C** *and answer* ***four*** *questions only.*
(*Each question carries* **150** *marks.*)

Part B

1. (i) Explain the following statement, with examples.
"Soft technology is always simple and cost-effective."
(ii) Describe the impact of wind and humidity on the agricultural production in Sri Lanka.
(iii) Describe how different plant parts are processed to make medicinal products giving appropriate examples.

2. (i) Explain the important activities in relation to proper management of a business.
(ii) Describe the risk of using chemical disinfectants at home to sanitize fresh fruits and vegetables giving appropriate examples.
(iii) Explain the importance of propagation by seeds in food crop production.

3. (i) Describe the importance of food processing in achieving national level food security.
(ii) Explain how the fishery industry contributes to the pollution of aquatic ecosystems.
(iii) Explain the following statement.
"Use of some of the modern biotechnological techniques increases the need for preservation of genetic resources."

Part C

4. (i) Giving **two** examples, describe the advantages of using appropriate technology in Sri Lankan agriculture.
(ii) Write the important components of main garden types in landscape architecture.
(iii) Describe the main steps involved in making good quality grass silage.

5. (i) Describe various methods of preservation of milk by sterilization.
(ii) Explain why pesticides have become popular for pest control in agriculture.
(iii) Describe the factors to be considered in application of chemical fertilizer for a newly established brinjal crop in the Dry zone.

6. (i) Describe the purposes of using misters in protected culture.
(ii) Explain the correct procedure for preparing a growth medium for establishing Gerbera as a cut flower crop for the export market.
(iii) Compare the advantages and disadvantages of artificial incubation and natural incubation of eggs.

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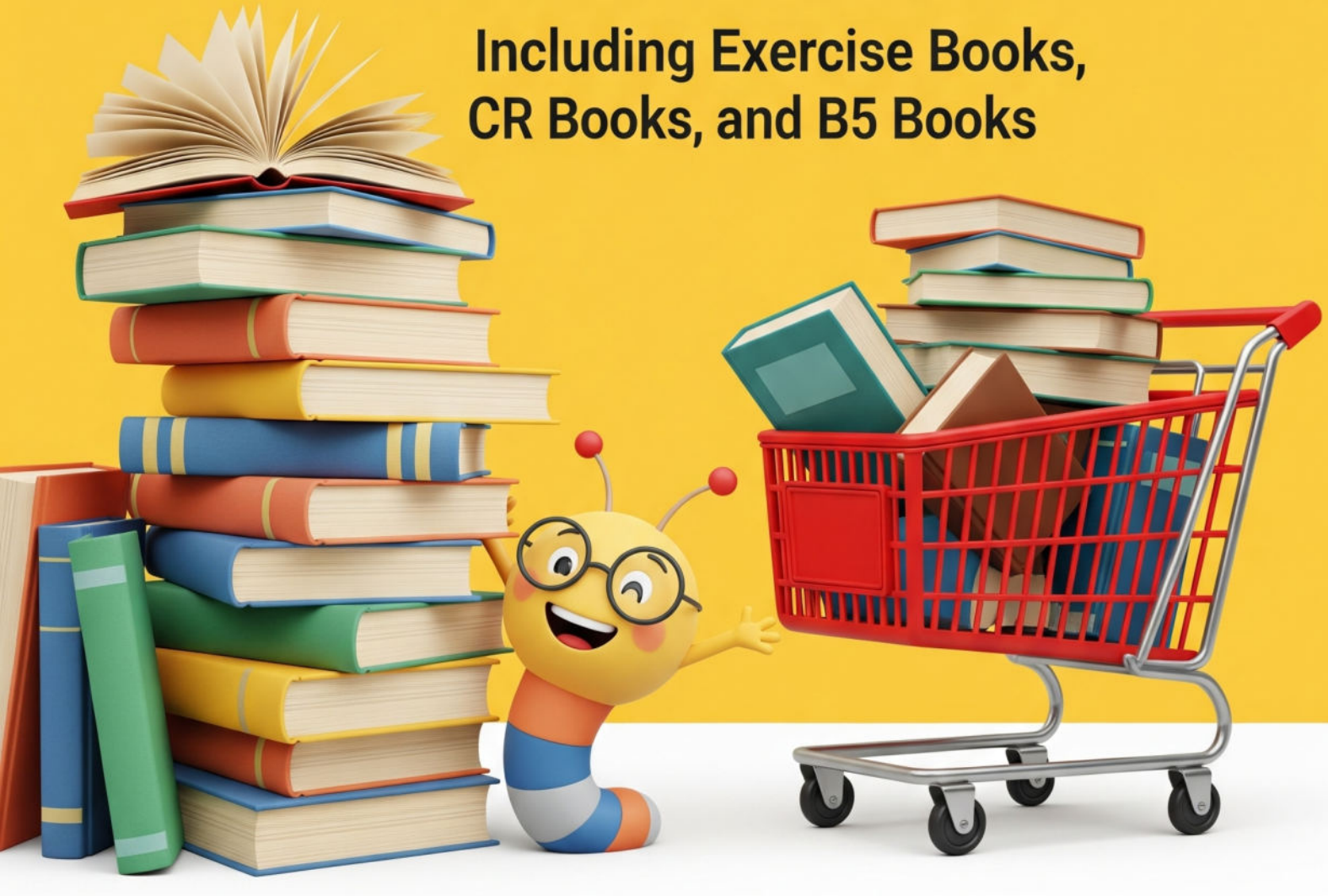
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